

ACUTE LIMB ISCHEMIA (ALI)

Definition

- **Sudden reduction in arterial perfusion** threatening limb viability — native artery, graft, or stent. **Symptoms <2 weeks** (>2 wks = CLTI). Incidence ~1/6000/year.

Etiology

1. **Arterial embolism (cardiac origin — AF, mural thrombus, ventricular aneurysm).**
2. **Acute-on-chronic thrombosis** (on pre-existing stenosis).
3. Thrombosis of bypass graft/stent.
4. Traumatic arterial injury.
5. Peripheral aneurysm thrombosis (esp **popliteal**).
6. Acute dissection.

Pathophysiology

- Abrupt flow interruption → hypoxia + metabolic acidosis. **Sequence: sensory dysfunction → motor paralysis → muscle necrosis. Reversible window ~6 hours.**

Sources of emboli

- **Cardiac: AF (left atrial appendage thrombus)**, mural thrombus (post-MI), prosthetic valves, paradoxical embolism (PFO).
- **Non-cardiac:** atheroembolism (aortic arch debris).

Mortality/morbidity

- Emergency diagnosis crucial. **Mortality 9–22%; limb salvage 70–90% if treated in first 6 hours.** Worse after **thrombotic vs embolic** occlusion.

Clinical — the "6 Ps"

- **Pain** (sharp, continuous, not relieved by analgesics), **Pallor**, **Pulselessness**, **Poikilothermia (cold)**, **Paraesthesia** (sensory nerve), **Paralysis** (late, motor nerve damage).
- Not all required for diagnosis.

Rutherford classification (ALI)

| Grade | Category | Sensory | Motor | Doppler (A/V) | Prognosis |

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| I | Viable | None | None | Audible / audible | No immediate threat |

| IIA | Marginally threatened | Minimal (toes) | None | Inaudible / audible | Salvageable if prompt |

| IIB | Immediately threatened | >toes | Mild-mod | Inaudible / audible | **Salvageable with immediate revascularization** |

| III | Irreversible | Profound anaesthesia | Paralysis | Inaudible / inaudible | **Major tissue loss inevitable** |

Investigations

- Don't delay revascularization (within 6h). Baseline: CBC, renal, coagulation, **CK**, electrolytes. **Mostly clinical diagnosis.** Duplex US, CTA, **DSA (gold standard — diagnose + treat).**

Management

- **Initial: systemic heparin (UFH bolus 100 IU/kg + infusion, aPTT 2–3×)**, analgesia (IV/PCA, not IM), hydration + O₂, monitor urine/renal (**myoglobinuria**).
- **Revascularization:**
- **Endovascular:** catheter-directed thrombolysis (tPA — Rutherford I-IIA), mechanical thrombectomy.
- **Open: Fogarty catheter embolectomy (embolic, normal artery);** bypass/endarterectomy (thrombosed atherosclerotic); hybrid.
- **Reperfusion complications: compartment syndrome, rhabdomyolysis (hyperkalaemia, acidosis), myoglobinuria → AKI.**

ALI due to arterial trauma

- **Hard signs (one = diagnostic, go to surgery):** pulsatile bleeding, absent distal pulses, expanding/pulsatile haematoma, bruit/thrill, distal ischaemia.
- **Soft signs (investigate):** diminished pulses, small stable haematoma, associated fracture, vessel-course injury, unexplained hypotension.
- **Management:** ATLS first; haemorrhage control (pressure/shunt/tourniquet); repair (primary/end-to-end, interposition vein graft, bypass); **fasciotomy** for compartment syndrome.

High-yield one-liners

- **ALI = <2 weeks; 6-hour reversible window; mortality 9–22%.**
- **Embolism = cardiac (AF); thrombotic worse prognosis.**
- **6 Ps; Rutherford I-III; IIB = immediate revascularization, III = amputation.**
- **Heparin immediately; Fogarty embolectomy (embolic); thrombolysis (I-IIA).**
- **Reperfusion → compartment syndrome + rhabdomyolysis + AKI; fasciotomy.**
- **Trauma: one hard sign = surgery; soft signs = investigate.**

ACUTE LIMB ISCHEMIA (ALI) — Revision Table

Bold = high-yield.

DEFINITION & ETIOLOGY	
Topic	Key points
Definition	Sudden ↓arterial perfusion; <2 weeks (>2wks = CLTI); 6-hour reversible window
Etiology	Embolism (cardiac — AF, mural thrombus) , acute-on-chronic thrombosis, graft thrombosis, trauma, aneurysm
Emboli sources	Cardiac: AF (LA appendage), post-MI mural thrombus , prosthetic valve, paradoxical (PFO)
Mortality	9–22%; limb salvage 70–90% if <6h ; thrombotic worse than embolic

CLINICAL & CLASSIFICATION	
Topic	Key points
6 Ps	Pain, Pallor, Pulselessness, Poikilothermia, Paraesthesia, Paralysis (late)
Sequence	Sensory → motor → muscle necrosis
Rutherford	I viable; IIA marginal (toes); IIB immediate threat (revascularize NOW); III irreversible (amputation)
Investigations	Mostly clinical; CK; DSA gold standard ; don't delay revascularization

MANAGEMENT	
Topic	Key points
Initial	Systemic heparin (UFH 100 IU/kg) , IV analgesia, hydration, monitor renal (myoglobinuria)
Endovascular	Thrombolysis (tPA) for I-IIA , mechanical thrombectomy
Open	Fogarty embolectomy (embolic, normal artery) ; bypass/endarterectomy (thrombotic)
Reperfusion complications	Compartment syndrome, rhabdomyolysis (hyperK, acidosis), myoglobinuria → AKI
Arterial trauma	Hard signs (1 = surgery): pulsatile bleed, absent pulse, expanding haematoma, bruit/thrill ; soft signs = investigate; ATLS; fasciotomy